



Lahbib NAIMI

Associate Professor
Artificial Intelligence & Software Engineering

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 lahbib-naimi



Profile

PhD in Computer Science, Associate Professor at the École Supérieure de Technologie de Meknès. Specialized in the **integration of artificial intelligence** into the development of smart tourism applications, **Model-Driven Architecture/Engineering (MDA/MDE)**, and **automatic code generation**. My goal is to contribute to higher education and applied research in software engineering and AI.



Academic Education

PhD in Computer Science

2020–2025

Faculty of Sciences and Techniques of Errachidia – Moulay Ismail University, Morocco

Specialty: Software Engineering and Artificial Intelligence

Thesis: *A Model-Driven and AI-Assisted Framework to Accelerate the Development of Tourism Software Applications*

Distinction: Highest honors with congratulations of the jury

Master's in Sciences and Techniques

2018–2020

FST Errachidia – Moulay Ismail University, Morocco

Specialty: Software Engineering and Information Systems (ILSI)

Distinction: Very Good — 2nd in class

Thesis: *Towards a software factory for web applications.*

Bachelor's in Sciences and Techniques

2016–2017

FST Errachidia – Moulay Ismail University, Morocco

Specialty: Software Engineering — 2nd in class

Project: *Development of a medical clinic management application.*

Bachelor's Degree in English Studies

2022–2025

Multidisciplinary Faculty – Moulay Ismail University, Morocco

Distinction: Fairly Good

Program focused on literature, linguistics, and academic communication in English.

Higher Technician Certificate (BTS)

2009–2011

Technical High School of Errachidia, Morocco

Field: Computer Science and New Information and Communication Technologies (INTIC)

Top of class

Baccalaureate in Physical Sciences

2009

Moulay Ali Cherif High School, Er-Rich, Morocco

Track: Physical Sciences

Distinction: Good



Professional and Academic Experience

Higher Education

Associate Professor — École Supérieure de Technologie de Meknès

2025–Present

Programs: Computer Engineering & Artificial Intelligence

Duties: Teaching, supervision, and coordination of courses in the *Computer Engineering* and *Artificial Intelligence* programs.

Lectures, tutorials, and practical work covering the fundamentals of software engineering and artificial intelligence techniques.

Adjunct Lecturer — Multidisciplinary Faculty of Errachidia

2025–2026

Semester 3

Module: UML

Lectures and tutorials on object-oriented modeling and software design with UML.

Adjunct Lecturer — Multidisciplinary Faculty of Errachidia

2025–2026

Master's — Semester 3

Module: Digitalization and Sharia Science: Application and Implementation

Course on the integration of digital tools in Sharia sciences and their practical applications.

Adjunct Lecturer — FST Errachidia

2024–2025

Master ILSI — Semester 3

Module: Introduction to Scientific Research Methodology and Innovation

Hours: 50 h

Lectures and tutorials on the principles of scientific research, academic writing, and technological innovation.

Adjunct Lecturer — FST Errachidia

2024–2025

Master ILSI — Semester 3

Half-module: Web Services

Hours: 28 h

Introduction to service-oriented architectures and distributed web technologies.

Adjunct Lecturer — FST Errachidia

2022

Bachelor's in Sciences and Techniques / S3 MIP — Tutorials

Tutorials: Algorithms and Programming

Hours: 56 h

Supervision of students in problem-solving and implementation of structured programs.

Secondary/Middle School Education

Computer Science Teacher

2012–2015

Moulay Ismail Douira Middle School, Aoufous

Subject: Computer Science | **Level:** Middle School

Computer Science Teacher

2015–Present

Abi Salim El Ayachi Middle School, Er-Rich

Subject: Computer Science | **Level:** Middle School

Professional Internships

Mobile Development Internship — Richadev

Oct. 2023 – Mar. 2024

Errachidia, Morocco

Project: Development of mobile applications for management and client interaction

Skills: User interface, usability, functional testing

Participation in the full development cycle and functional validation.

Web Design Internship — Deconait

Apr. 2011

Salé, Morocco

Project: Design of a dynamic website and promotional materials

Skills: Digital creation, graphic design, visual communication

Introduction to the web design process and digital layout.



International Training Programs and Study Visits

AI Awareness and Technology Integration in Education

July 16–29, 2025

NITTTR Chennai, India

International program organized by the **Ministry of Education, Government of India**.

Participation as a Moroccan teacher-researcher within the ITEC program.

KLIC LEADer 2024

August 2024

Seoul, Republic of Korea

Theme: *Digital Education Leadership and Innovation*

Intensive training focused on digital transformation, instructional design, and educational system governance.

KLIC Educators 2023

May 2023

Jeollanam-do, Republic of Korea

Training program in digital education and academic leadership.

Participation in collaborative workshops on the use of AI and emerging technologies in education.



Academic Supervision

- **Supervision of capstone projects (Master ILSI – FST Errachidia)**

Supervision of several Master's students in the design and development of web and mobile applications. Topics: software engineering, applied AI, recommendation tools for smart tourism.

- **Supervision of professional bachelor's projects (FST Errachidia)**

Guidance of students on UML modeling, databases, and dynamic interfaces.

- **Academic mentoring and technical support**

Assistance with technical report writing and preparation for oral defenses. Methodological support for planning and documentation.

- **Participation in thesis defense evaluation**

Member of examination boards (bachelor's and master's): technical, methodological, and scientific communication assessment.



Research Project

A Software Factory for Developing Intelligent Tourism Mobile Applications

Research structure: Software Engineering and Information Systems Engineering Group (GL-ISI) & Research Laboratory in Software Engineering and Artificial Intelligence (GLIA)

Partner institutions: Faculty of Sciences and Techniques – Errachidia, Moulay Ismail University – Meknès, and Université du Québec à Rimouski (UQAR)

Objective: Development of a software factory integrating Artificial Intelligence and Model-Driven Engineering to automate the creation of smart tourism applications.



Scientific Contributions

Articles published in Scopus-indexed international journals

1. **Naimi, L.**, Benaddi, L., Ouaddi, C., Souha, A., Abdelmalek, H., & Jakimi, A. (2025). *Model-Driven Architecture for Accelerating the Development of Tourism Applications*. IEEE Access.
2. **Naimi, L.**, Ouaddi, C., Benaddi, L., et al. (2025). *Machine Learning Approach to Identify Promising*

Mountain Hiking Destinations Using GIS and Remote Sensing. International Journal of Advanced Computer Science & Applications, 15(10).

3. **Naimi, L.**, Benaddi, L., Jakimi, A., et al. (2024). *Assessing Habitat Suitability for Aoudad (*Ammotragus lervia*) Reintroduction in Southeastern Morocco to Promote Ecotourism*. Scientific African, 26, e02444.
4. **Naimi, L.**, Manaouch, M., & Jakimi, A. (2024). *Leveraging Machine Learning Algorithms to Identify Potential Geosites for Geotourism Promotion in Ziz Upper Watershed in Southeastern Morocco*. GeoJournal of Tourism and Geosites, 57, 2041–2051.
5. Ouaddi, C., Benaddi, L., **Naimi, L.**, et al. (2025). *Assessing the Effectiveness of Large Language Models for Intent Detection in Tourism Chatbots: A Comparative Analysis and Performance Evaluation*. Scientific African, 28, e02649.
6. Souha, A., Ouaddi, C., Benaddi, L., **Naimi, L.**, et al. (2025). *MDE-Based Approach for Accelerating the Development of Recommender Systems in Smart Tourism*. IEEE Access.
7. Manaouch, M., **Naimi, L.**, Jakimi, A., et al. (2025). *Enhancing Geotourism in Southeastern Morocco through Machine Learning-Based Geomorphosite Identification*. Geoheritage, 17(1), 1–19.

Papers presented at Scopus-indexed international conferences

1. **Naimi, L.**, Abdelmalek, H., & Jakimi, A. (2024). *A DSL-Based Approach for Code Generation and Navigation Process Management in a Single Page Application*. Procedia Computer Science, 231, 299–304.
2. **Naimi, L.**, Jakimi, A., Saadane, R., et al. (2024). *Automating Software Documentation: Employing LLMs for Precise Use Case Description*. Procedia Computer Science, 246, 1346–1354.
3. **Naimi, L.**, Souha, A., Benaddi, L., et al. (2024). *Large Language Models for Decision-Making Enhancement in Tourism Package Customization*. In: DASA 2024. IEEE, pp. 1–5.
4. **Naimi, L.**, Manaouch, M., Jakimi, A., et al. (2024). *A New Approach for Automatic Test Case Generation from Use Case Diagram Using LLMs and Prompt Engineering*. In: ICCSC 2024. IEEE, pp. 1–5.
5. **Naimi, L.**, Bouziane, E. M., & Jakimi, A. (2024). *A Model-Driven Approach for Developing Smart Tourism Web Applications*. In: ICDATA 2024. Springer, pp. 255–263.
6. **Naimi, L.**, & Jakimi, A. (2023). *Toward an Automating Web User Interface Development Using a DSL-Based Approach*. In: ICATH 2023. Springer, pp. 273–277.
7. **Naimi, L.**, Jakimi, A., El Bermi, L., et al. (2024). *A Sketch of a MDA Approach to Accelerate the Development of User Interfaces in Tourism Web Applications*. In: MSCC 2024. IEEE, pp. 1–5.
8. **Naimi, L.**, Ouaddi, C., Benaddi, L., et al. (2025). *Towards a Domain Model of Smart Tourism Based on the 6As Framework*. In: SCA 2025. Springer, pp. 278–287.
9. Souha, A., **Naimi, L.**, Ouaddi, C., et al. (2024). *Towards a Software Factory for Developing Recommender Systems and Decision Support Systems for Smart Tourism*. In: DASA 2024. IEEE, pp. 1–4.
10. Bouziane, E. M., **Naimi, L.**, Ouaddi, C., et al. (2024). *Use Cases and Scenarios Engineering in Smart Software Application Development Services*. In: CommNet 2024. IEEE, pp. 1–7.
11. Abdelmalek, H., **Naimi, L.**, Khriss, I., et al. (2023). *A Sketch of a Model-Driven Approach to Develop Smart Tourism Mobile Applications*. In: CSMF 2023. MDPI, p. 2.
12. Ouaddi, C., Benaddi, L., **Naimi, L.**, et al. (2024). *A Graphical DSL for Accelerating Chatbot Development*. In: CommNet 2024. IEEE, pp. 1–6.
13. Benaddi, L., Ouaddi, C., **Naimi, L.**, et al. (2024). *A Decision Support System for Enhancing Transportation Services Using Aspect-Based Sentiment Analysis*. In: DASA 2024. IEEE, pp. 1–5.
14. Ouaddi, C., Benaddi, L., Souha, A., **Naimi, L.**, et al. (2024). *A Metamodel for a DSL to Design Decision Support Systems in Smart Tourism*. In: DASA 2024. IEEE, pp. 1–5.
15. Bouziane, E. M., Abdelmalek, H., Benaddi, L., Ouaddi, C., **Naimi, L.**, et al. (2024). *UML Functional Aspects in Business-to-Business: Some Lessons Learned*. In: ESAI 2024. IEEE, pp. 1–6.

Non-indexed international conferences

1. **Naimi, L.**, Bouziane, E. M., & Jakimi, A. (2024). *Automating Test Case Generation from Class Diagram Using Generative AI*. In: ICSMAI 2024. Springer, pp. 133–140.

2. **Naimi, L.**, Benaddi, L., Ouaddi, C., Souha, A., & Jakimi, A. (2024). *A Software Factory for Mobile Applications in Smart Tourism*. In: ICCSA 2024, Errachidia, Morocco.
3. **Naimi, L.**, & Jakimi, A. (2021). *Towards a software factory for web applications*. In: JDOCST'21, Errachidia, Morocco.
4. **Naimi, L.**, & Jakimi, A. (2023). *Generative AI: An Introduction and Application with ChatGPT*. In: Language and AI Conference, Errachidia, Morocco.
5. Jakimi, A., **Naimi, L.**, et al. (2021). *Towards a software factory for the development of mobile applications for smart tourism*. In: 88th Congrès de l'Acfas, UQAR, Canada.
6. Jakimi, A., Ouaddi, C., Benaddi, L., **Naimi, L.**, & Souha, A. (2024). *Artificial Intelligence and Tourism in Morocco: Challenges and Opportunities*. LASMITI Seminar, Rabat, Morocco.

Work in Progress

1. *Automated Code Generation from UML Class Diagrams Using Large Language Models and Few-Shot Learning*. (First author). Article under review at **Automated Software Engineering (Springer)**.
2. *Automating Eco-Friendly Instruction Generation in Tourism Applications Using Large Language Models*. (First author). Article accepted for publication at the **International Conference on Sustainable Energy and Intelligent Systems (ICSEIS 2025)**.



Scientific Events and Training

Involvement in Organizing Scientific Events

- **The 4th International Conference on Computer Science and its Applications (ICSA'2024)**, Errachidia, Morocco — December 2024.
- **2nd edition of the Scientific Days on Smart Technologies & Innovation (J2STI'23)**, FST Errachidia — May 2023.
- **3rd International Day on Computer Science and Applied Mathematics (ICSAM'23)**, FST Errachidia — May 2023.
- **1st edition of the Scientific Days on Smart Technologies & Innovation: AI & Quantum Computing (J2STI'22)**, FST Errachidia — May 2022.

Training and Seminars (Elsevier, Web of Science, Scopus)

- **Elsevier / Scopus Workshops**
Participation in several training sessions on literature search, scientific publishing, and author profile management:
 - Finding Relevant Articles and Selecting Journals for Publication
 - Managing Scopus Profiles & Author Visibility
 - Avoiding Predatory Journals & Research Ethics
 - Research Related to Sustainable Development Goals
 - Essential Literature Review with Scopus
 - Exporting Data from Scopus & Analyzing Research Productivity
 - Everything About Author Profiles (Scopus Perspective)
 - Scopus Ready 2023
- **Web of Science / Clarivate Trainings**
Specialized training on scientific publishing, bibliometrics, and academic visibility:
 - Publishing in a Reputable Journal
 - Selecting the Right Journal for Publication
 - Analyze Research Impact with InCites
 - Organizing Bibliography with EndNote Online & Journal Citation Reports

– Open Science and Research Transparency

- **IMIST & UMI Doctoral Workshops**

Doctoral seminars on research methodology, literature review, scientific communication, and conceptual model design:

- Research Methodology and Scientific Communication
- Systematic Literature Review with Zotero & NVIVO
- Digital Identity of Researcher & Bibliographic Management
- Epistemological Approaches and Research Frameworks



Associations and Extra-Academic Activities

- **Executive Member – National Association of Computer Science Teachers**

Contribution to educational initiatives and programs promoting computer science education nationwide.

- **Participation in national competitions — e.g., finals of the Inwi Challenge**

Development of innovative educational games using *Minecraft Education*.



Languages

Arabic: Fluent, used in professional communication.

French: Advanced, proficient in scientific writing and communication.

English: Good academic and research level.

Holder of a *Bachelor's Degree in English Studies* (Moulay Ismail University, 2025).



Awards and Honors

In recognition of my academic performance, I graduated **top of my class** in the *Higher Technician Certificate (BTS)*.

I also achieved **2nd place** in my *Bachelor's in Computer Science* as well as in my *Master's in Software Engineering and Information Systems*.

Finally, I defended my *PhD thesis in Computer Science* with the distinction of **Highest Honors with Congratulations of the Jury**.



Interests

Digital transformation of education • Pedagogical innovation • Applied artificial intelligence • Regional development • Scientific culture